

Polarization Models for Atomistic Simulation: From Drude Oscillators to Self-Consistent Field Methods

Formation Engine Development Team

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Abstract

Accurate representation of electronic polarization is essential for modeling molecular systems with strong electrostatic interactions, including hydrogen bonds, metal complexes, and crystalline materials. This document evaluates three approaches to incorporating polarization into atomistic force fields: (1) explicit Drude particle methods, (2) complex multi-component atomic representations with orbital structure, and (3) self-consistent field (SCF) induced dipole models. We analyze each method’s theoretical foundation, computational cost, numerical stability, and compatibility with deterministic simulation frameworks. Our analysis demonstrates that SCF-based induced dipole models provide the optimal balance of accuracy, stability, and computational efficiency for the Formation Engine platform, offering 10–30% accuracy improvements for hydrogen-bonded systems and 50%+ for metal complexes with minimal integration complexity.

1 Introduction

1.1 Motivation: The Polarization Problem

Classical molecular mechanics force fields traditionally employ fixed point charges to represent electrostatic interactions. While computationally efficient, this approximation fails to capture the dynamic response of electron distributions to local electric fields—a phenomenon critical for:

- **Hydrogen bonding networks:** Polarization strengthens H-bonds by 10–30% relative to fixed-charge models [?].
- **Metal–ligand coordination:** d-orbital back-bonding and charge transfer require explicit polarization for accurate binding energies [?].
- **-stacking interactions:** Induced dipole–induced dipole interactions dominate aromatic stacking geometries [?].
- **Dielectric response:** Crystalline materials exhibit directional polarizability essential for electro-optical properties.

The Formation Engine currently implements fixed-charge Coulomb interactions (Section ??), achieving stable energy conservation but underestimating polar interaction strengths. This document establishes the theoretical framework for extending the platform to include explicit polarization effects.

1.2 Scope and Objectives

We evaluate three distinct approaches to polarization modeling:

1. **Drude oscillators** (Section ??): Explicit massless or massive "electron" particles harmonically bound to nuclear centers.
2. **Composite atomic structures** (Section ??): Multi-component representations including nucleon cores and orbital rings/shells.
3. **Self-consistent field (SCF) induced dipoles** (Section ??): Implicit polarization via field-dependent induced dipoles computed iteratively.

For each method, we analyze:

- **Physical fidelity:** Ability to reproduce quantum mechanical polarization responses.
- **Computational cost:** Scaling behavior and integration timestep constraints.
- **Numerical stability:** Sensitivity to integration parameters and field strengths.
- **Deterministic reproducibility:** Compatibility with bit-exact trajectory replication.

Our conclusion (Section ??) recommends SCF-based induced dipoles as the optimal path forward, with detailed implementation guidance in Section ??.

2 Current Implementation: Fixed-Charge Coulomb Model

The Formation Engine presently models electrostatic interactions via the Coulomb potential with fixed atomic charges:

$$U_{\text{Coul}}(\{\mathbf{r}_i\}) = \frac{1}{4\pi\epsilon_0} \sum_{i < j} \frac{q_i q_j}{r_{ij}} \quad (1)$$

where q_i are static charges assigned via:

- **Charge Equilibration (QEq)** [?]: Pre-simulation minimization of electronegativity differences.
- **Fixed partial charges:** Manually assigned or loaded from parameterized force fields.

Implementation Details:

- Location: `atomistic/models/lj_coulomb.cpp`
- Combining rules: Lorentz–Berthelot for LJ, direct Coulomb for charges
- Cutoff: Quintic switching function ($0.9r_c \rightarrow r_c$)
- Periodic boundaries: Minimum image convention (orthogonal boxes)

2.1 Limitations of Fixed Charges

Physical inadequacy: Electron distributions deform in response to local electric fields $\mathbf{E}(\mathbf{r})$, inducing dipoles $\boldsymbol{\mu} = \boldsymbol{\alpha} \cdot \mathbf{E}$ where $\boldsymbol{\alpha}$ is the atomic polarizability tensor. Fixed charges cannot capture:

1. **Field-dependent energy stabilization:** $U_{\text{pol}} = -\frac{1}{2}\boldsymbol{\mu} \cdot \mathbf{E}$
2. **Cooperative many-body effects:** Induced dipoles create secondary fields, requiring self-consistent iteration.
3. **Directional anisotropy:** Hybridized orbitals (e.g., sp^2 , sp^3) exhibit tensor polarizability.

Quantitative errors:

- H-bond strengths: 10–30% underestimated [?]
- Metal–ligand binding: 50–100% errors in coordination energies [?]
- Cohesive energies: 5–10% errors in dense phases [?]

Consequence for Formation Engine: Crystal structure predictions, reactive site identification (Fukui functions), and thermodynamic property calculations (e.g., dielectric constants) are systematically biased toward under-polarized configurations.

3 Method 1: Drude Oscillator Model

3.1 Theoretical Formulation

The Drude model represents atomic polarization by attaching a fictitious charged particle (the "Drude oscillator") to each polarizable atom via a harmonic spring [?, ?]:

$$U_{\text{Drude}} = \frac{1}{2}k_D(\mathbf{r}_D - \mathbf{r}_c)^2 \quad (2)$$

where:

- $\mathbf{r}_c$: Position of the atomic core (charge q_c)
- $\mathbf{r}_D$: Position of the Drude particle (charge $q_D < 0$)
- k_D : Spring constant (typically 500–1000 kcal/mol/Å²)

The induced dipole emerges from the displacement:

$$\boldsymbol{\mu} = q_D(\mathbf{r}_D - \mathbf{r}_c) \quad (3)$$

Under an applied field $\mathbf{E}$, equilibrium is achieved when the electric force balances the spring restoring force:

$$q_D\mathbf{E} = k_D(\mathbf{r}_D - \mathbf{r}_c) \quad \Rightarrow \quad \boldsymbol{\mu} = \frac{q_D^2}{k_D}\mathbf{E} \equiv \boldsymbol{\alpha}\mathbf{E} \quad (4)$$

thus defining the atomic polarizability $\alpha = q_D^2/k_D$.

3.2 Integration Schemes

Two approaches exist for propagating Drude particles:

3.2.1 Massive Drude (Explicit Dynamics)

Assign the Drude particle a small mass m_D (typically 0.1–0.4 amu) and integrate its equations of motion explicitly:

$$m_D \ddot{\mathbf{r}}_D = q_D \mathbf{E}_{\text{ext}} - k_D (\mathbf{r}_D - \mathbf{r}_c) \quad (5)$$

Advantages:

- Naturally time-dependent polarization response (relevant for optical properties).
- No linear system solve required.
- Compatible with velocity Verlet integrators.

Critical Challenges:

1. **Stiff dynamics:** The spring frequency $\omega_D = \sqrt{k_D/m_D} \approx 1000 \text{ cm}^{-1}$ requires timesteps $\Delta t \ll 2\pi/\omega_D \approx 0.1 \text{ fs}$ (vs. 1 fs for typical MD).
2. **Thermostat coupling:** Drude oscillators at temperature $T_D \approx 1 \text{ K}$ prevent kinetic energy theft from physical degrees of freedom. Requires dual thermostats (core + Drude) [?].
3. **Numerical instability:** Strong fields or close contacts can cause Drude "fly-away" unless constrained.

3.2.2 Massless Drude (SCF-Drude)

Set $m_D \rightarrow 0$ and enforce instantaneous equilibrium via iterative relaxation at each MD step:

$$\mathbf{r}_D^{(n+1)} = \mathbf{r}_c + \frac{q_D}{k_D} \mathbf{E}^{(n)} \quad (6)$$

where $\mathbf{E}^{(n)}$ includes contributions from all other Drude dipoles (self-consistent iteration).

Advantages:

- Removes high-frequency modes (allows standard timesteps).
- No thermostat coupling issues.

Disadvantages:

- Requires iterative linear solve (similar cost to direct SCF methods).
- Loses time-dependent polarization response.

3.3 Evaluation

Verdict: Drude oscillators are well-suited for applications requiring explicit polarization dynamics (e.g., spectroscopy, ultrafast phenomena). For equilibrium properties and crystal structure optimization, the added complexity is unjustified relative to direct SCF methods.

Table 1: Drude Oscillator Performance Assessment

Criterion	Assessment
Physical fidelity	High: Captures anisotropy via directional springs, time-dependent response.
Computational cost	High: 2–3 $\times$ slower (massive) or iterative solve (massless).
Stability	Medium: Requires careful damping/constraints to prevent fly-away.
Determinism	Medium: Stiff dynamics amplify floating-point sensitivity.
Extensibility	Medium: Adding anisotropy or frequency-dependent response is straightforward but increases complexity.

[Figure 1: Schematic of Drude oscillator model showing atomic core, Drude particle, spring constant, and induced dipole vector.]

Figure 1: Drude oscillator representation. Each polarizable atom (core) is attached to a charged Drude particle via a harmonic spring. Displacement under an applied field generates an induced dipole moment.

4 Method 2: Composite Atomic Structures with Orbital Rings

4.1 Conceptual Framework

This approach replaces point atoms with structured composite objects comprising:

- **Nucleon core:** Homogeneous mass distribution representing the nucleus.
- **Orbital rings/shells:** Spring-connected pseudo-particles representing electron density at orbital radii (e.g., 2s, 2p shells).

Polarization emerges from deformation of the orbital structure under external fields.

4.2 Theoretical Motivation

Appealing features:

1. **Explicit orbital anisotropy:** Directional orbital arrangements (e.g., p_x , p_y , p_z) naturally yield tensor polarizability.
2. **Hybridization representation:** Orbital ring geometry can encode sp^2/sp^3 character.

3. **Pedagogical clarity:** Visually interpretable atomic structure.

4.3 Critical Implementation Challenges

4.3.1 Parameter Explosion

Each atom requires specification of:

- Number of orbital shells (1 for H, 2–3 for second-row elements, etc.)
- Ring radii (r_{shell})
- Spring constants (radial k_r , tangential k_t , inter-shell k_{ij})
- Charge distributions ($q_{\text{core}}, q_{\text{shell}}$)
- Angular constraints (orbital directionality)

Problem: For 20 elements, with 3 shells per atom and 5 parameters per shell, this yields $20 \times 3 \times 5 = 300$ parameters to calibrate, with no established parameterization protocol.

4.3.2 Stiff Coupled Oscillators

The system becomes a dense network of coupled harmonic oscillators with widely varying frequencies:

$$m_i \ddot{\mathbf{r}}_i = \sum_{j \in \text{neighbors}} k_{ij} (\mathbf{r}_j - \mathbf{r}_i - \mathbf{r}_{ij}^0) \quad (7)$$

Consequence: Numerical stiffness far exceeding Drude models. Timesteps must be reduced to $\Delta t < 0.01$ fs to maintain stability.

4.3.3 Constraint Drift

Orbital rings require geometric constraints (e.g., maintaining spherical/ellipsoidal symmetry, angular momentum conservation). Constraint violation accumulates exponentially:

$$|\mathbf{C}(t)| \sim e^{\lambda t} \quad (8)$$

where λ is the largest Lyapunov exponent of the constrained system.

4.3.4 Wrong Physical Scale

Nuclear structure (nucleons) is irrelevant for chemical bonding. Including nuclear degrees of freedom introduces:

- Artificial coupling between nuclear and electronic timescales.
- Computational cost with zero chemical accuracy benefit.

4.4 Evaluation

Verdict: This approach is unsuitable for production use. It may have pedagogical value as a conceptual toy model but lacks a feasible path to accurate parameterization or stable integration.

Table 2: Composite Atomic Structure Assessment

Criterion	Assessment
Physical fidelity	Unknown: No established parameterization; likely uncalibratable.
Computational cost	Catastrophic: 10–100× slower due to stiffness and constraint solving.
Stability	Poor: Severe constraint drift, resonance instabilities.
Determinism	Poor: Chaotic sensitivity to timestep and rounding errors.
Extensibility	Low: Adding elements requires inventing new orbital geometries.

[Figure 2: Schematic of composite atomic structure showing nucleon core surrounded by orbital ring structures representing electron shells.]

Figure 2: Composite atomic structure with orbital rings. Each atom comprises a central nucleon core and concentric orbital shells connected by springs. While conceptually appealing, this representation introduces severe numerical stiffness and parameter calibration challenges.

5 Method 3: Self-Consistent Field (SCF) Induced Dipole Model

5.1 Theoretical Foundation

The SCF approach treats polarization implicitly by assigning each atom an induced dipole moment $\boldsymbol{\mu}_i$ that depends self-consistently on the local electric field [?, ?]:

$$\boldsymbol{\mu}_i = \boldsymbol{\alpha}_i \cdot \mathbf{E}_i^{\text{loc}} \quad (9)$$

where $\boldsymbol{\alpha}_i$ is the atomic polarizability tensor (isotropic: $\boldsymbol{\alpha}_i = \alpha_i \mathbf{I}$) and the local field includes contributions from:

1. Permanent charges: $\mathbf{E}_i^{\text{perm}} = \sum_{j \neq i} \frac{q_j \mathbf{r}_{ij}}{r_{ij}^3}$
2. Other induced dipoles: $\mathbf{E}_i^{\text{ind}} = \sum_{j \neq i} \mathbf{T}_{ij} \cdot \boldsymbol{\mu}_j$

The dipole–dipole interaction tensor is:

$$\mathbf{T}_{ij} = \frac{1}{r_{ij}^3} \left(\mathbf{I} - 3 \frac{\mathbf{r}_{ij} \otimes \mathbf{r}_{ij}}{r_{ij}^2} \right) \quad (10)$$

Combining these yields the self-consistent equation:

$$\boldsymbol{\mu}_i = \alpha_i \left(\mathbf{E}_i^{\text{perm}} + \sum_{j \neq i} \mathbf{T}_{ij} \cdot \boldsymbol{\mu}_j \right) \quad (11)$$

This is a $3N \times 3N$ linear system (3 dipole components per atom):

$$(\mathbf{I} - \mathbf{A}) \boldsymbol{\mu} = \mathbf{E}^{\text{perm}} \quad (12)$$

where $\mathbf{A}_{ij} = \alpha_i \mathbf{T}_{ij}$ for $i \neq j$ and $\mathbf{A}_{ii} = 0$.

5.2 Computational Solution Strategies

5.2.1 Direct Linear Solve

For small systems ($N < 500$), use direct LU decomposition:

$$\boldsymbol{\mu} = (\mathbf{I} - \mathbf{A})^{-1} \mathbf{E}^{\text{perm}} \quad (13)$$

Cost: $O(N^3)$ for matrix inversion, $O(N^2)$ per subsequent solve.

5.2.2 Iterative Damped Relaxation

For larger systems, use iterative update with damping:

$$\boldsymbol{\mu}_i^{(k+1)} = (1 - \omega) \boldsymbol{\mu}_i^{(k)} + \omega \alpha_i \left(\mathbf{E}_i^{\text{perm}} + \sum_{j \neq i} \mathbf{T}_{ij} \cdot \boldsymbol{\mu}_j^{(k)} \right) \quad (14)$$

$$\omega \approx 0.3\text{--}0.7 \quad (\text{damping factor}) \quad (15)$$

Convergence criterion:

$$\max_i \|\boldsymbol{\mu}_i^{(k+1)} - \boldsymbol{\mu}_i^{(k)}\| < 10^{-6} \text{ Debye} \quad (16)$$

Typical convergence: 5–20 iterations.

Cost: $O(N^2)$ per iteration.

5.2.3 Preconditioned Conjugate Gradient (PCG)

For very large systems ($N > 5000$), use PCG with neighbor-list sparsity:

$$\mathbf{A}_{ij} = 0 \quad \text{for} \quad r_{ij} > r_c \quad (17)$$

Cost: $O(N)$ per iteration with spatial decomposition.

5.3 Damping Functions: Thole Screening

At short range ($r_{ij} < 2 \text{ \AA}$), the point-dipole approximation breaks down, leading to "polarization catastrophe" (infinite induced dipoles). Thole introduced an empirical damping function [?]:

$$\mathbf{T}_{ij}^{\text{damp}} = \mathbf{T}_{ij} \cdot f_{\text{damp}}(u) \quad (18)$$

where $u = r_{ij}/(\alpha_i \alpha_j)^{1/6}$ and:

$$f_{\text{damp}}(u) = 1 - \left(1 + \frac{au}{2}\right) e^{-au} \quad (19)$$

with $a = 2.6$ (empirically optimized).

Physical interpretation: Damping represents electron cloud overlap, preventing unphysical short-range polarization.

5.4 Energy and Forces

The polarization energy is:

$$U_{\text{pol}} = -\frac{1}{2} \sum_i \boldsymbol{\mu}_i \cdot \mathbf{E}_i^{\text{perm}} \quad (20)$$

Forces on atom i arise from:

$$\mathbf{F}_i^{\text{pol}} = -\nabla_i U_{\text{pol}} \quad (21)$$

$$= -\sum_j q_j \left(\frac{\mathbf{r}_{ij}}{r_{ij}^3} (\boldsymbol{\mu}_i \cdot \mathbf{r}_{ij}) + \frac{\boldsymbol{\mu}_i}{r_{ij}^3} - 3 \frac{(\boldsymbol{\mu}_i \cdot \mathbf{r}_{ij}) \mathbf{r}_{ij}}{r_{ij}^5} \right) \quad (22)$$

$$- \sum_j (\nabla_i \mathbf{T}_{ij}) : (\boldsymbol{\mu}_i \otimes \boldsymbol{\mu}_j) \quad (23)$$

5.5 Evaluation

Table 3: SCF Induced Dipole Model Assessment

Criterion	Assessment
Physical fidelity	High: Reproduces QM polarization energies within 5–10% for equilibrium configurations.
Computational cost	Low–Medium: $O(N^2)$ per MD step (iterative) or $O(N)$ (sparse).
Stability	Excellent: Thole damping prevents polarization catastrophe; linear system is well-conditioned.
Determinism	Excellent: Fixed convergence criterion ensures bit-exact reproducibility.
Extensibility	High: Straightforward to add tensor $\boldsymbol{\alpha}_i$, multipoles, frequency dependence.

Verdict: SCF-based induced dipoles provide the optimal balance of accuracy, stability, and computational efficiency. This is the recommended approach for the Formation Engine.

[Figure 3: Flowchart of SCF iteration showing: (1) Compute permanent field, (2) Iterative dipole update, (3) Convergence check, (4) Force evaluation.]

Figure 3: SCF induced dipole iteration workflow. At each MD timestep, induced dipoles are computed via iterative relaxation until self-consistency. The converged dipoles then contribute to energy and force evaluations.

6 Comparative Analysis

Table 4: Comprehensive Method Comparison

Property	Fixed Charge	Drude	Composite	SCF
Physical fidelity	Low	High	Unknown	High
Computational cost	$1\times$	$2\text{--}3\times$	$10\text{--}100\times$	$1.5\text{--}2\times$
Stability	Excellent	Medium	Poor	Excellent
Determinism	Excellent	Medium	Poor	Excellent
Timestep	1 fs	0.1 fs (massive)	<0.01 fs	1 fs
Parameters per atom	1 (q)	3 (q_c, q_D, k_D)	15+	2 (q, α)
Implementation complexity	Low	Medium	High	Medium
Extensibility	Low	Medium	Low	High
Accuracy gain (H-bonds)	—	+15–30%	?	+10–30%
Accuracy gain (metals)	—	+50%	?	+50%

6.1 Decision Matrix

For the Formation Engine’s objectives (crystal structure prediction, reactive site identification, multiscale property extraction), the critical factors are:

1. **Deterministic reproducibility:** Essential for validation and debugging $\Rightarrow$ Excludes massive Drude, composite.
2. **Computational efficiency:** Must scale to $N = 10^4\text{--}10^6$ atoms $\Rightarrow$ Excludes composite.
3. **Stability:** Long-timescale simulations (ns– μ s) $\Rightarrow$ Requires robust integration.
4. **Accuracy:** 10–30% improvement needed for H-bonding, metal complexes $\Rightarrow$ Requires explicit polarization.

Conclusion: SCF-based induced dipoles satisfy all constraints with minimal integration risk.

7 Implementation Roadmap for Formation Engine

7.1 Phase 1: Minimal SCF Core (1–2 weeks)

7.1.1 Data Structures

Extend `atomistic::State` to include:

- `std::vector<Vec3> induced_dipoles` (per-atom μ_i)
- `std::vector<double> polarizabilities` (per-atom α_i)

7.1.2 Parameter Assignment

Use literature values for polarizabilities [?]:

- H: $0.667 \text{ \AA}^3$
- C: $1.76 \text{ \AA}^3$
- N: $1.10 \text{ \AA}^3$
- O: $0.802 \text{ \AA}^3$
- S: $2.90 \text{ \AA}^3$
- Metals: $4\text{--}20 \text{ \AA}^3$ (element-dependent)

7.1.3 Core Algorithm

File: `atomistic/models/scf_polarization.hpp`

```
void compute_induced_dipoles(State& s, double tol = 1e-6) {
    // 1. Compute permanent electric field at each atom
    std::vector<Vec3> E_perm = compute_permanent_field(s);

    // 2. Iterative SCF loop
    for (int iter = 0; iter < max_iter; ++iter) {
        std::vector<Vec3> mu_new(s.N);

        for (uint32_t i = 0; i < s.N; ++i) {
            // Field from other induced dipoles
            Vec3 E_ind = compute_induced_field(s, i);

            // Update dipole with damping
            mu_new[i] = s.polarizabilities[i] * (E_perm[i] + E_ind);
            mu_new[i] = 0.5 * (s.induced_dipoles[i] + mu_new[i]);
        }

        // Check convergence
    }
}
```

```

        if (dipole_converged(s.induced_dipoles, mu_new, tol)) break;

        s.induced_dipoles = mu_new;
    }
}

```

7.2 Phase 2: Thole Damping (1 week)

Add short-range damping to `compute_induced_field()`:

```

double u = r_ij / pow(alpha_i * alpha_j, 1.0/6.0);
double damp = 1.0 - (1.0 + 0.5*a*u) * exp(-a*u); // a = 2.6
Vec3 T_ij = compute_dipole_tensor(r_ij) * damp;

```

7.3 Phase 3: Energy and Forces (1–2 weeks)

Implement:

- `double eval_polarization_energy(State& s)`
- `void add_polarization_forces(State& s)`

Integrate into `atomistic/models/lj_coulomb.cpp` after fixed-charge Coulomb evaluation.

7.4 Phase 4: Validation (2–3 weeks)

Benchmark against:

1. **Water clusters:** Compare to AMOEBA force field results [?].
2. **Metal complexes:** Reproduce coordination energies within 10% of DFT.
3. **Crystalline ice:** Validate dielectric constant ($\epsilon_r \approx 3.2$ at 250 K).

7.5 Phase 5: Performance Optimization (1–2 weeks)

- Implement neighbor-list-based sparse matrix-vector products.
- Add preconditioned conjugate gradient for $N > 5000$.
- GPU acceleration (optional: CUDA kernel for dipole updates).

7.6 Phase 6: Anisotropic Extension (Optional, 2–3 weeks)

Replace scalar α_i with tensor α_i (3×3):

- Assign based on local bonding geometry (sp^2 : planar, sp^3 : tetrahedral).
- Useful for layered materials (graphene, MoS_2).

8 Conclusion

This document has rigorously evaluated three approaches to incorporating electronic polarization into atomistic force fields:

1. **Drude oscillators:** Physically accurate but numerically stiff; suitable only for applications requiring explicit polarization dynamics.
2. **Composite atomic structures:** Conceptually appealing but computationally intractable; lacks feasible parameterization pathway.
3. **SCF induced dipoles:** Optimal balance of accuracy, stability, and efficiency; recommended for implementation.

The SCF approach offers:

- **10–30% accuracy improvements** for hydrogen-bonded systems.
- **50%+ improvements** for metal–ligand complexes.
- **Deterministic reproducibility** with fixed convergence criteria.
- **Computational cost** of 1.5–2 $\times$ relative to fixed-charge models.
- **Natural extensibility** to tensor polarizabilities and multipole moments.

The proposed 6-phase implementation plan provides a clear path forward, with Phase 1–3 deliverables achievable within 4–5 weeks. This extension will elevate the Formation Engine’s predictive accuracy to a level competitive with state-of-the-art polarizable force fields (AMOEBA, Drude-2013) while maintaining the platform’s commitment to deterministic, reproducible simulation.

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